GEOMETRIC IRREDUCIBILITY CAN BE TESTED OVER A SEPARABLE ALGEBRAIC CLOSURE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement carries the chapter's slogan environment, rendered by the no-op display environment defined at lines 75-77 of the retained preamble exactly as the author wrote it.

Editorial note. The counted spans carry 3 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-characterize-geometrically-irreducible is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Irreducible components of a base change to a separable algebraic closure. Same chapter, Lemma lemma-separably-closed-field-irreducible-components, invoked at proof lines 1563-1564. Supplies that irreducibility is preserved by extending the base field to a larger separably closed field.
- Criterion for irreducibility of a scheme via an affine open covering. Properties, Lemma properties-lemma-characterize-irreducible, invoked at proof lines 1575-1578. Supplies that a reducible member or an empty intersection of two members makes X reducible.
- Geometric irreducibility of a finite type algebra tested over a finite separable extension. Algebra, Lemma algebra-lemma-geometrically-irreducible, invoked at proof lines 1583-1586. Supplies the finite separable extension over which an affine member becomes reducible.

Source reference. Original lemma 037K: sources/stacks-chow/tags.txt, source lines 4203–4203. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 038H: sources/stacks-varieties/pinned-original-source.tex, source lines 1524–1524. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 01OM: sources/stacks-chow/tags.txt, source lines 2199–2199. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Geometric irreducibility can be tested over a separable algebraic closure of the base field. Let k be a field. Let X be a scheme over k. The following are equivalent:*

1. *X is geometrically irreducible over k,*
2. *for every finite separable field extension k'/k the scheme $X_{k'}$ is irreducible, and*
3. *$X_{\overline{k}}$ is irreducible, where $k \subset \overline{k}$ is a separable algebraic closure of k.*

Proof. Assume $X_{\overline{k}}$ is irreducible, i.e., assume (3). Let k'/k be a field extension. There exists a field extension $\overline{k}'/\overline{k}$ such that k' embeds into $\overline{k}'$ as an extension of k. By Lemma 038H we see that $X_{\overline{k}'}$ is irreducible. Since $X_{\overline{k}'} \to X_{k'}$ is surjective we conclude that $X_{k'}$ is irreducible. Hence (1) holds.

Let $k \subset \overline{k}$ be a separable algebraic closure of k. Assume not (3), i.e., assume $X_{\overline{k}}$ is reducible. Our goal is to show that also $X_{k'}$ is reducible for some finite subextension $\overline{k}/k'/k$. Let $X = \bigcup_{i \in I} U_i$ be an affine open covering with U_i not empty. If for some i the scheme U_i is reducible, or if for some pair $i \neq j$ the intersection $U_i \cap U_j$ is empty, then X is reducible (Properties, Lemma 01OM) and we are done. In particular we may assume that $U_{i,\overline{k}} \cap U_{j,\overline{k}}$ for all $i, j \in I$ is nonempty and we conclude that $U_{i,\overline{k}}$ has to be reducible for some i. According to Algebra, Lemma 037K this means that $U_{i,k'}$ is reducible for some finite separable field extension k'/k. Hence also $X_{k'}$ is reducible. Thus we see that (2) implies (3).

The implication (1) $\Rightarrow$ (2) is immediate. This proves the lemma. $\square$